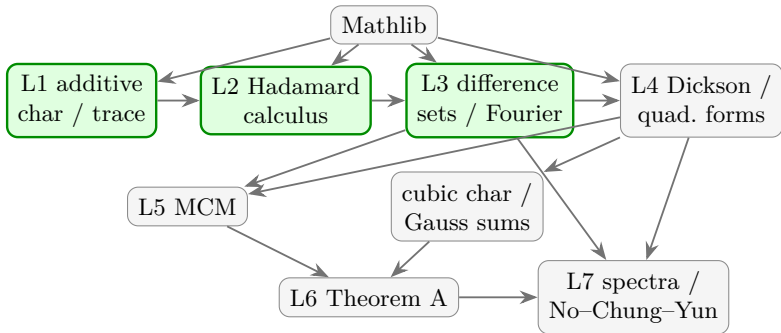
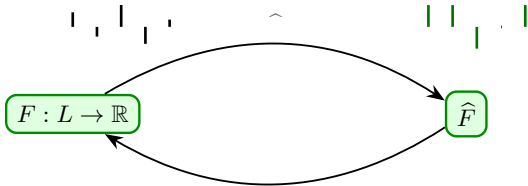


Layering (import DAG, rooted in Mathlib)



edges = import; a cycle would be a compile error

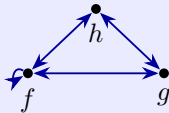
Duality $\hat{}$: an involution



$$\hat{\hat{F}} = F \quad (\text{self-dual pairing; Parseval } \langle F, G \rangle = \langle \hat{F}, \hat{G} \rangle)$$

Equivalence (quadratic forms; $\text{Arf} = \text{invariant}$)

$\text{Arf} = 0$ (O^+ , *hyperbolic*)

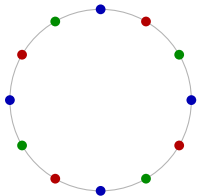


$\text{Arf} = 1$ (O^- , *elliptic*)



refl / symm / trans within a class; no arrow crosses colours (Arf preserved)

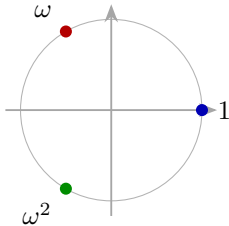
Isomorphism $L^*/(L^*)^3 \cong \mu_3 \subset \mathbb{C}$



$L^* = \langle g \rangle$, order $q - 1$

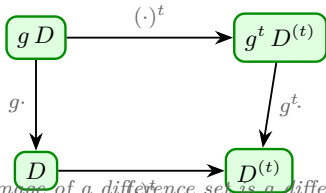
3 cube cosets

$$\chi_3 : x \mapsto x^{(q-1)/3}$$



$\mu_3 = \{1, \omega, \omega^2\}$

Multiplier (power map = automorphism of L^*)



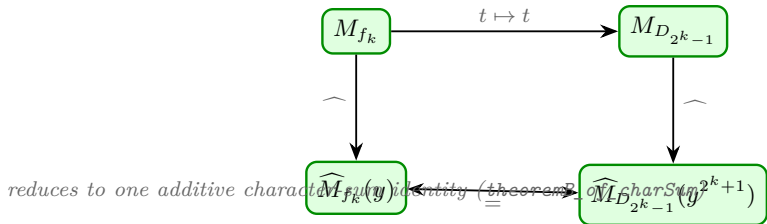
$\gcd(t, q-1) = 1 \Rightarrow (\cdot)^t$ bijective; image of a difference set is a difference set

Partition $\sum_v M_\eta(v) = q$

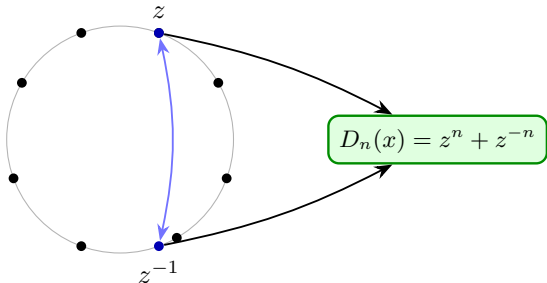


$$L = \bigsqcup_v \eta^{-1}(v) \quad (\text{a permutation} \iff \text{every block has size } 1)$$

Theorem B (transform dictionary = substitution)



Dickson folding $z \leftrightarrow z^{-1}$ (order $q+1$ torus)



$x = z + z^{-1}$; the two preimages fold together

permutation $\iff \gcd(n, q^2 - 1) = 1$

Theorem A route ($\approx, \hat{\approx}$ transitive)

